MIL-B-25820B(USAF)
6 April 1967
SUPERSEDING
MIL-B-25820A(USAF)
11 March 1960

MILITARY SPECIFICATION

BOMB, LOADING PRACTICE BDU-6/E AND BDU-35/E

1. SCOPE

1.1 This specification covers two types of equipment designated as Bomb, Loading Practice BDU-6/E and BDU-35/E. This item is nonexpendable and is not normally dropped from the carrying aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-L-31	Lacquer, Cellulose Nitrate, Gloss
TT-T-266	Thinner; Dope And Lacquer (Cellulose-Nitrate)
PPP-B-636	Box, Fiberboard
PPP-C-650	Crates, Wood, Open And Covered

Military

MIL-C-104	Crates, Wood; Lumber And Plywood Sheathed, Nailed And Bolted
MIL-P-116	Preservation, Methods Of
MIL-V-3083	Vibrator, Concrete, Pneumatic
MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aircraft And Missiles
MIL-W-27000	Weapons, Nuclear, Practice And Training - And Similar Special Weapons Items, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts

FSC 1L05

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MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Amchoring, Bracing, Blocking And Waterproofing; With Appropriate Test Methods

DRAWINGS

Air Force

58J45140	Bomb, Loading Practice BDU-6/E, Assembly
58J45141	Center Section Assembly
60D45077	Ballast Assembly
64E13139	Cable, Special Purpose Electrical, Branched
64J13142	Center Section Assembly (BDU-35/E)
64J13167	Bomb Assembly, Practice Loading, BDU-35/E

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The practice loading bombs shall consist of one long nose assembly or one short nose assembly, each fitting the same location pin and hole alignment; one center section assembly, welded to the aft case assembly; four small fins or four large fins or a combination of both four small and four large fins, with provisions provided for relocation or quantities in optional configurations; four interchangeable spoiler rings; and one aft cap section, as shown in Figure 1.

3.3 General specification. The requirements of MIL-W-27000 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.4 Design and construction. Design and construction of the BDU-6/E and BDU-35/E bombs shall be in accordance with Air Force Drawings 58J45140 and 64J13167 respectively, and all subsidiary detail drawings.

3.5 Concrete fill. The concrete fill process is to be accomplished by the contractor on all bombs.

3.5.1 Density. The concrete shall have a density of 143 pounds per cubic foot, plus 7, minus 18 pounds per cubic foot.

3.5.2 Recommended procedure for mixing. The recommended procedure for mixing the concrete is as specified in MIL-W-27000.

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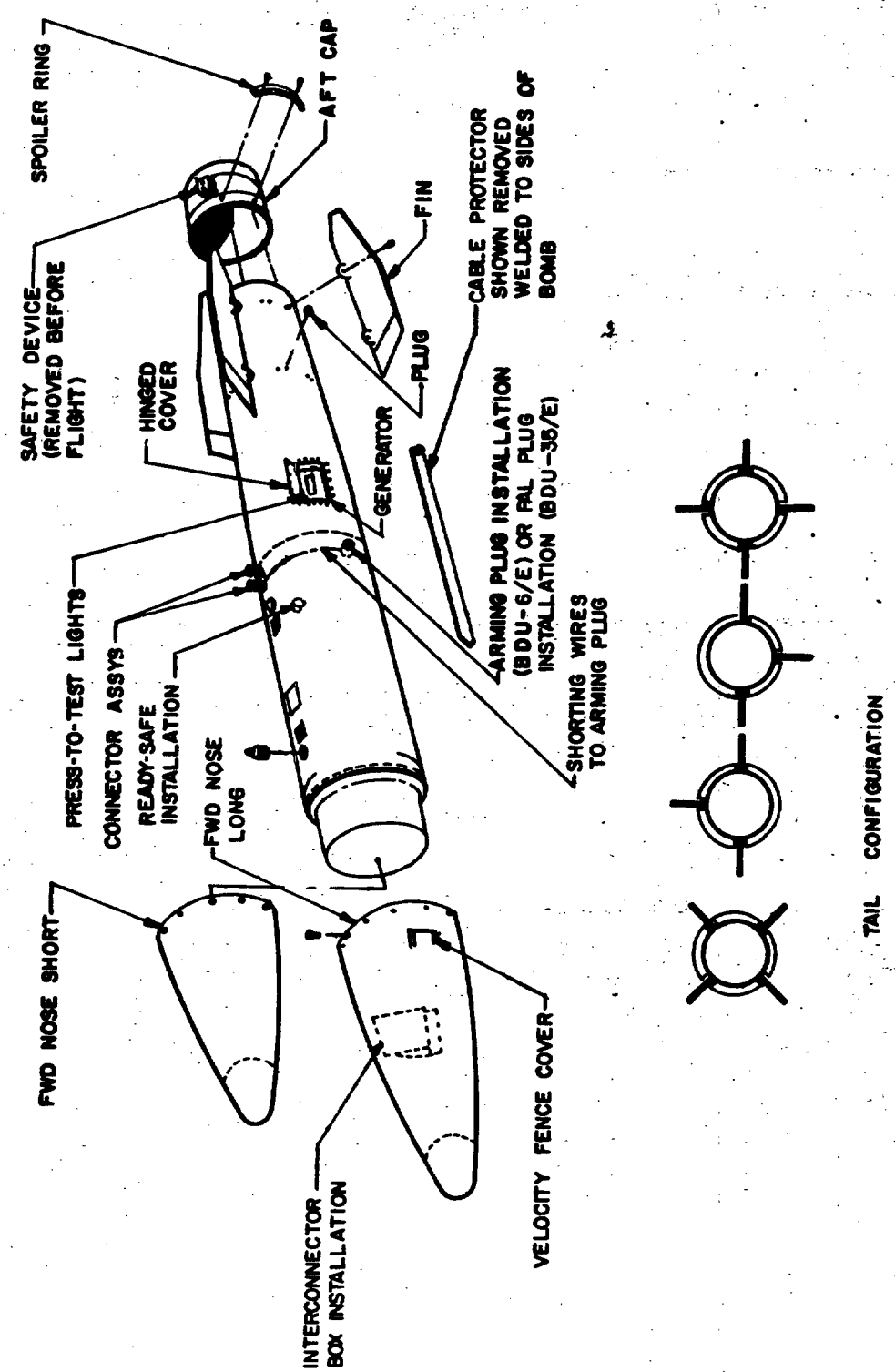


FIGURE 1. OVER-ALL VIEW OF BOMB, LOADING PRACTICE

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3.5.3 Method of filling with concrete. The bomb shall be filled with the concrete mixture in the following manner:

a. Before installing the ballast assembly, drawing 60D45077, the center section assembly, drawing 58J45141 (BDU-6/E) or drawing 64J13142 (BDU-35/E), shall be placed in a vertical position, aft end down, as shown in Figure 2.

b. The center section assembly shall be filled with concrete in increments of 1/4, 1/2, 3/4, and then to the full line as indicated on drawing 58J45141 (BDU-6/E) or drawing 64J13142 (BDU-35/E) and as shown in Figure 2.

c. Each fill increment shall be vibrated with a pneumatic concrete vibrator, MIL-V-3083, as shown in Figure 2, for from 30 to 60 seconds. The vibrator shall be removed between filling increments.

3.6 Performance.

3.6.1 Functional. Satisfactory functional performance of the bomb shall be interpreted as performance of simulated interconnector box and tester switch installations. (Reference Figure 3.)

3.6.2 Static loads. The bomb shall be capable of withstanding the static loads resulting from aborted ejection, fore and aft section moment and tail fin loading.

3.6.3 Environmental conditions (equipment nonoperational). The bomb shall not suffer damage, and shall satisfy its functional performance requirements after subjection to:

a. Shock loads as defined in MIL-W-27000.

b. Vibrations as defined in MIL-W-27000.

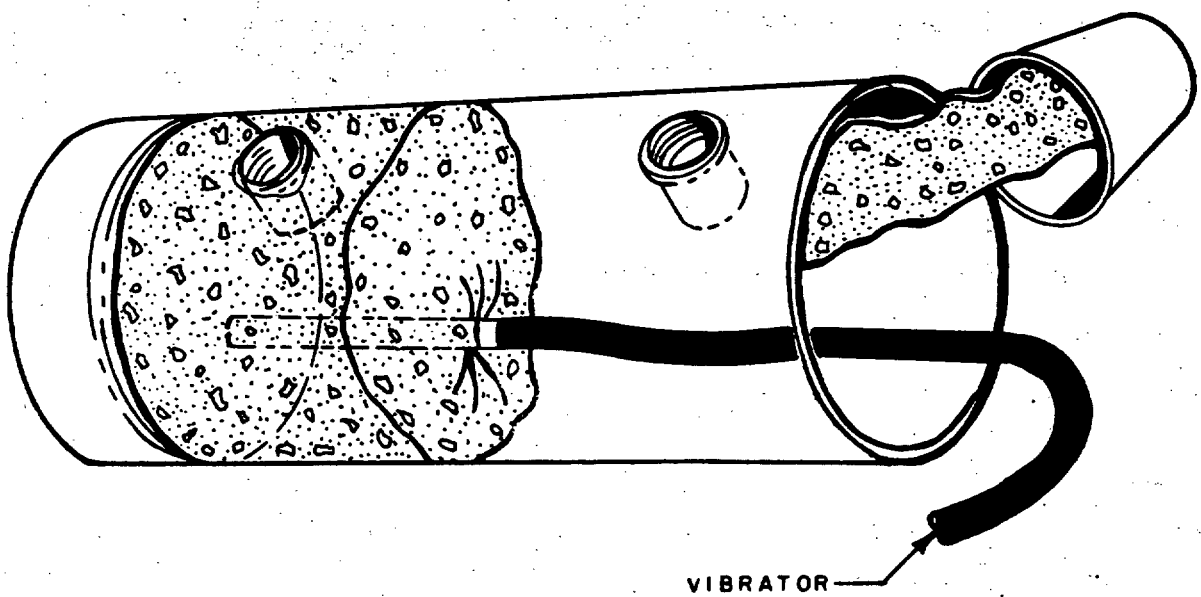
3.7 Interchangeability. Interchangeability requirements shall be in accordance with the provisions of MIL-I-8500. All long nose assemblies and short nose assemblies shall be interchangeable, fitting the same location pin and hole alignments on the center section assembly. All fins shall be interchangeable, and shall be capable of being relocated in any of the optional configurations, as shown in Figure 1. All four spoiler rings shall be interchangeable. All aft cap sections produced shall be interchangeable with aft case assemblies.

3.8 Finish process. The bomb shall be finished in accordance with the finish requirements of MIL-W-27000.

3.8.1 Finish coat on external parts of the bomb. All external metal parts shall be given a finish coat in accordance with the following:

a. Lacquer, gloss, in accordance with TT-L-31, color number 17875, white, shall be used for the finish coat.

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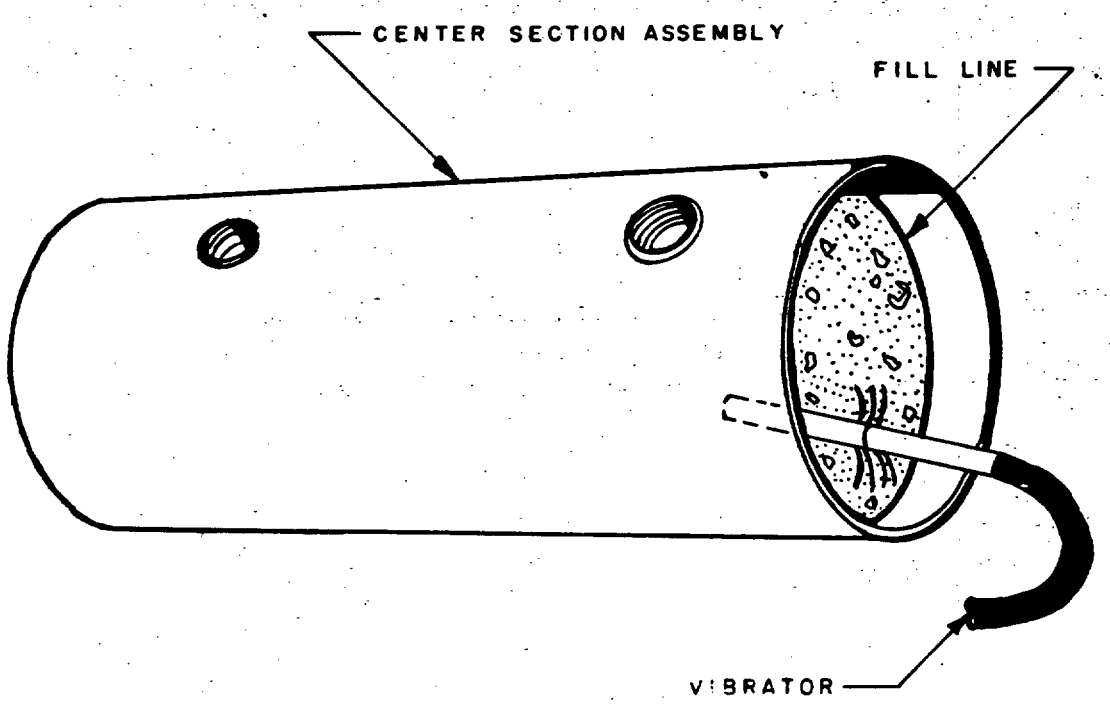


FIGURE 2. CENTER SECTION ASSEMBLY FILLED WITH CEMENT MIXTURE.

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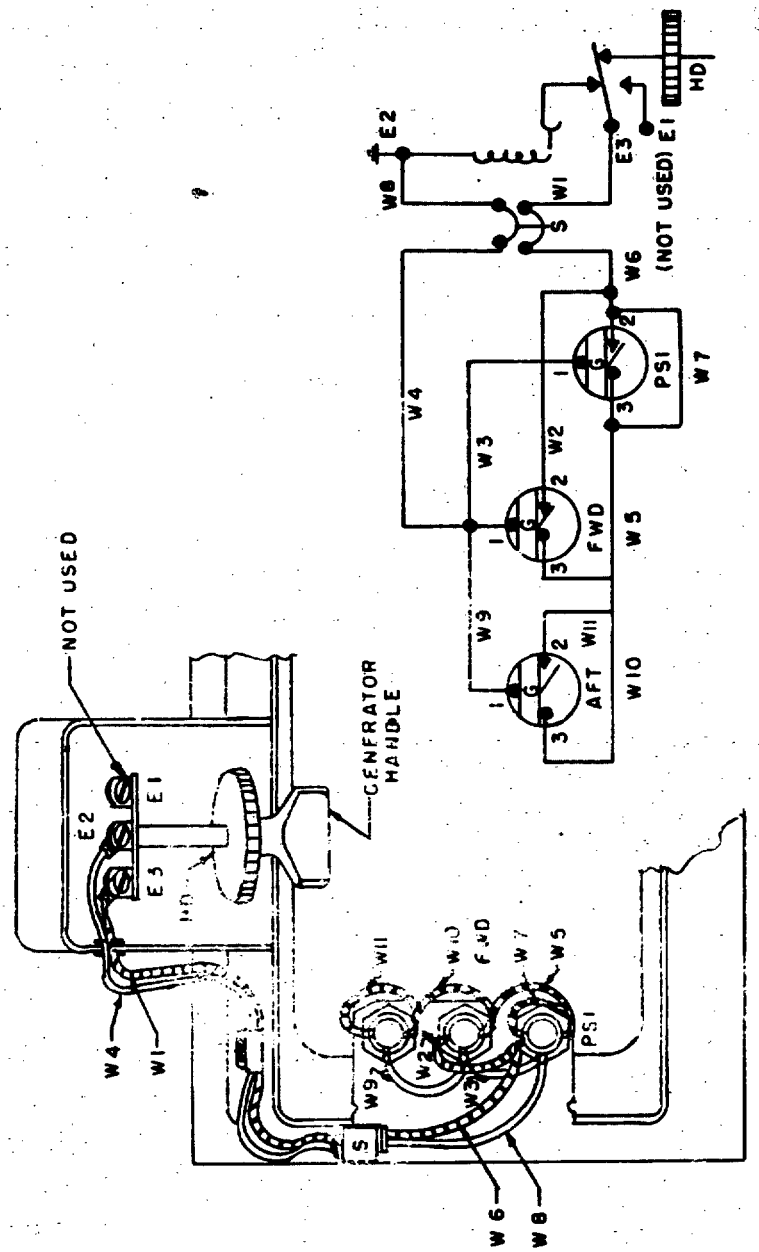


FIGURE 3 SCHEMATIC DIAGRAM AND PICTORIAL WIRING DIAGRAM

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b. The lacquer shall be thinned by one volume of lacquer thinner, TT-T-266, to two volumes of lacquer.

c. The lacquer shall be applied by spraying on a coating 0.5 to 1.0 mil in thickness.

d. Parts shall be allowed to air dry for a minimum of one hour prior to handling after receiving the finish coat.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Government furnished property. The following item of equipment shall be furnished to the contractor, upon his request, as Government furnished property, for inclusion in the bomb:

Item No.	Stock No.	Item Description	Quantity
1	5805-160-1032	Generator, Ringing, Hand, Type GN-38	1 each
2	5935-759-0397CM	Connector, Plug Electrical (SA1467)	1 each (BDU-35/E only)

3.11 Workmanship. Workmanship requirements shall be in accordance with MIL-W-27000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the bomb shall be classified as follows:

a. Preproduction testing . . . . . see 4.4.

b. Acceptance tests . . . . . see 4.5.

4.3 Test conditions. Test conditions are described under the test methods to which they apply.

4.4 Preproduction testing.

4.4.1 Preproduction test samples. The preproduction test units, prepared by the contractor, shall be representative of the production equipment, complete

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with concrete fill. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.4.1.1 Test report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish the report to the procuring activity.

4.4.2 Preproduction tests. Preproduction tests shall consist of all tests described in 4.6, and shall be arranged in the same order.

4.5 Acceptance tests. Acceptance tests shall consist of the following individual tests as described under 4.6.

a. Examination of product.

b. Conformance to drawings.

c. Functional test.

4.6 Test methods.

4.6.1 Examination of product. Each bomb shall be given a complete visual examination to determine compliance with the requirements of this specification, including workmanship.

4.6.2 Conformance to drawings. Each bomb shall be inspected for strict conformance to the engineering drawings and all specifications and standards referenced therein.

4.6.3 Functional test.

4.6.3.1 Test light circuit. A complete continuity test shall be conducted on each bomb to insure that all installations are properly wired and that each unit will function as intended. The schematic diagram and pictorial wiring diagram are shown in Figure 3. The diagram represents a circuit designed to light the PSI, FWD, and AFT press-to-test lamps whenever the hand-operated generator is turned.

4.6.3.2 Arming circuit (BDU-6/E). A continuity test shall be conducted on the arming circuit of each bomb. The wiring diagram of the arming circuit is shown in Figure 4.

4.6.3.3 PAL circuit (BDU-35/E). A continuity test shall be conducted on the PAL circuit of each BDU-35/E bomb. The wiring diagram of the PAL circuit is shown in Figure 5. The continuity of the PAL circuit shall be checked from pin to pin (J1, J2 and PAL receptacle P/N 65C33987) with an AN/PSM-6 Multimeter, or similar electrical test instrument, as follows:

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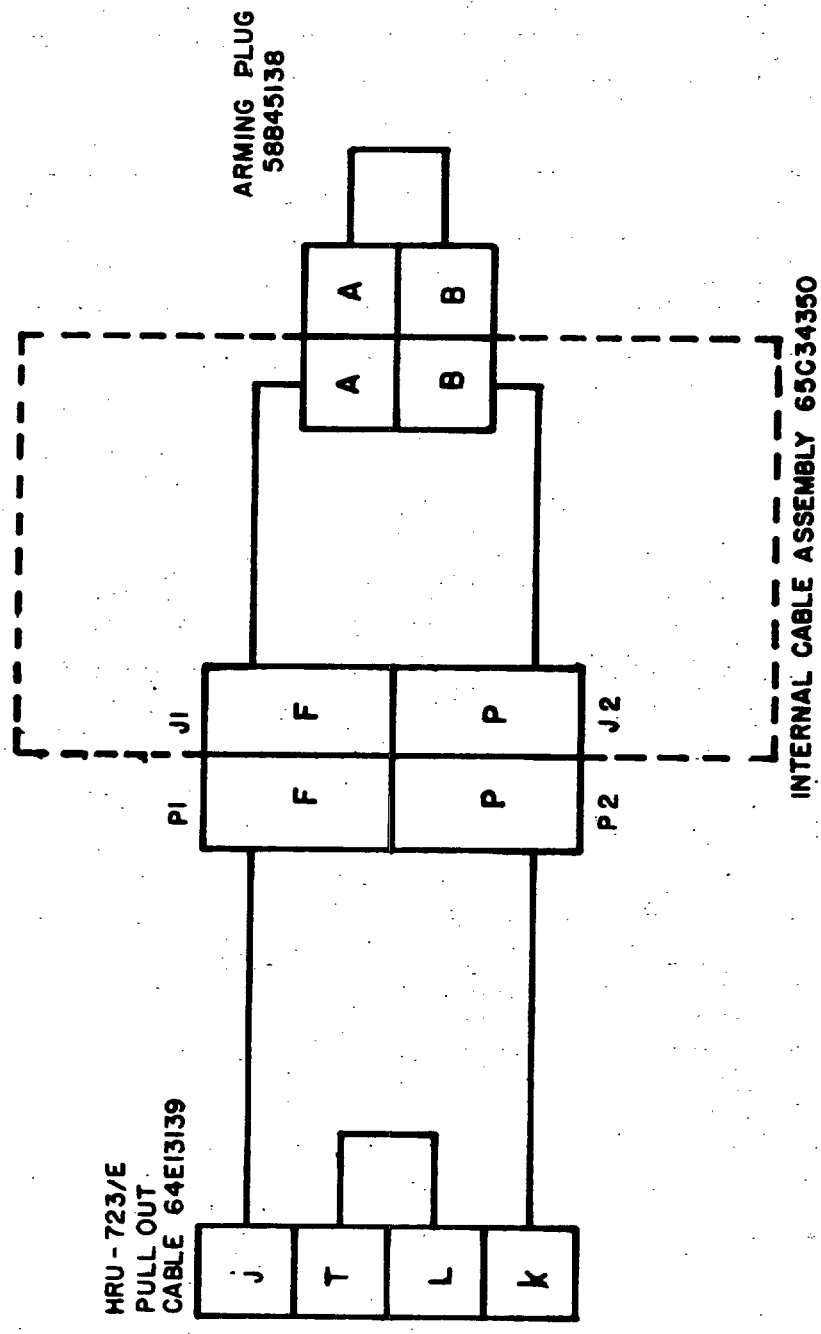


FIGURE 4 - WIRING DIAGRAM - BDU-6/E ARMING CIRCUIT

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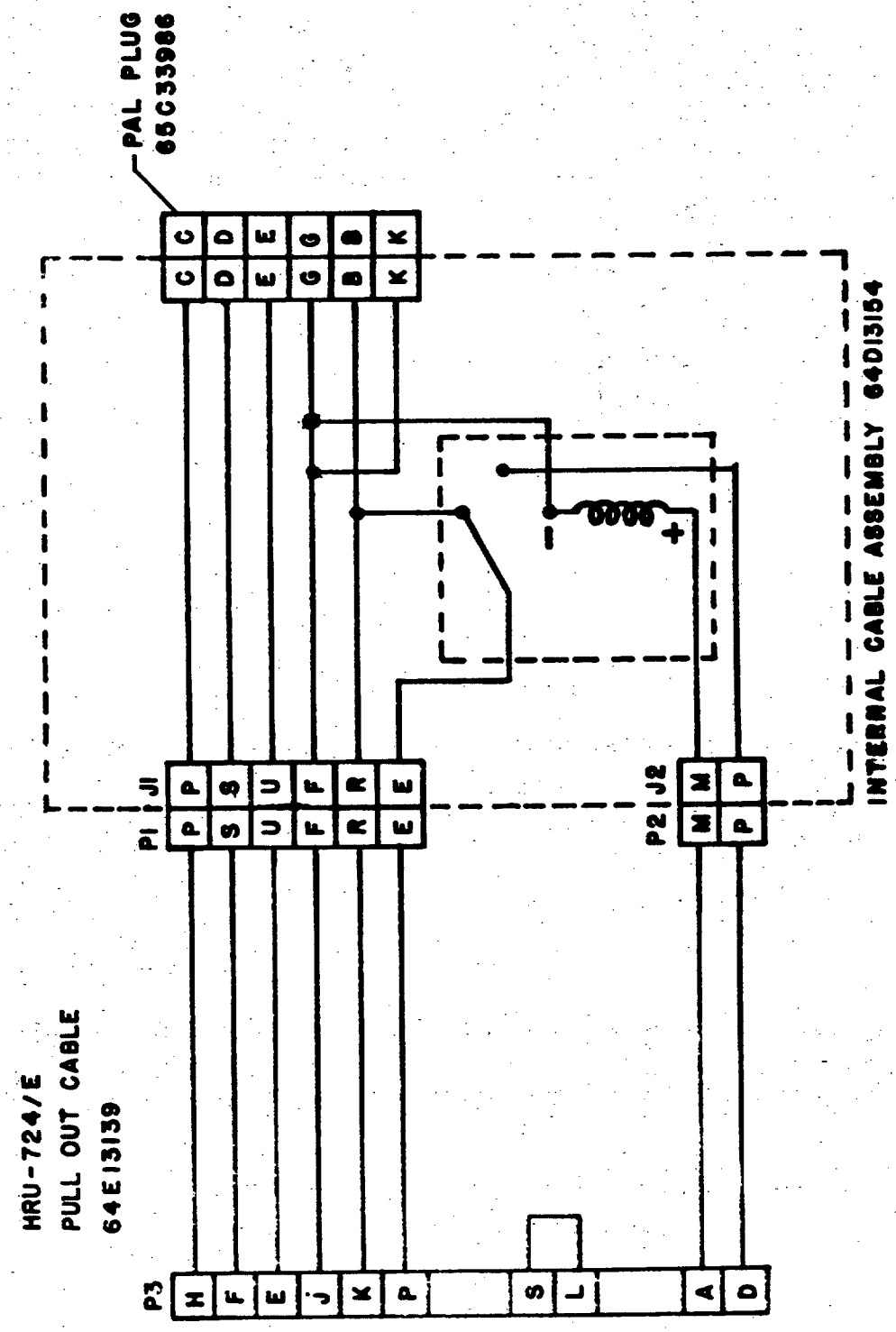


FIGURE 5 – WIRING DIAGRAM – BDU – 35/E PAL CIRCUIT

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From		To	
PIN	CONNECTOR	PIN	CONNECTOR
P	J1	C	PAL PLUG
S	J1	D	PAL PLUG
U	J1	E	PAL PLUG
F	J1	G	PAL PLUG
R	J1	B	PAL PLUG
F	J1	K	PAL PLUG
E	J1	B	PAL PLUG

(For the last check, apply 28 volts (v) direct current (dc) power between J2 Pin M (positive) and J1 Pin F (negative), and check continuity between J1 Pin E and J2 Pin P.)

4.6.3.4 Insulation resistance test (BDU-35/E). An insulation resistance test shall be conducted as specified under Method 302 of MIL-STD-202, Test Condition B. The potential shall be applied between the terminals indicated and the case, with the spare terminals indicated grounded to the case. The insulation resistance shall be a minimum of 100 megohms. Insulation resistance shall be checked as follows:

APPLY POTENTIAL BETWEEN CASE AND:	SPARE TERMINALS GROUNDED TO CASE:
J1 PIN P	J1 PINS S, U, F, R, & E J2 PINS M & P
J1 PIN S	J1 PINS U, F, R, & E J2 PINS M & P
J1 PIN U	J1 PINS F, R, & E J2 PINS M & P
J1 PIN R	J1 PIN F J2 PINS M & P
J2 PIN P	J1 PIN F J2 PIN M

(For the last insulation check, apply 28 v dc power between J2 Pin M (positive) and J1 Pin F (negative).)

J1 PIN E	J1 PINS R, U, S & P

4.6.4 Static load tests. These tests shall consist of:

a. Aborted Ejection Test (destruction).

b. Fore and Aft Section Moment Test (destruction).

c. Tail Fin Test (destruction).

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4.6.4.1 Test specimen preparation. Bombs to be subjected to static tests shall first be filled with concrete in accordance with the concrete fill requirements of this specification.

4.6.4.2 Reference axes. Reference axes and positive directions of those axes shall be as shown in Figure 6.

4.6.4.3 Aborted ejection test, Parts I and II.

4.6.4.3.1 Test set-up.

a. The bomb shall be suspended from a mock-up of suspension equipment attached to a test frame capable of withstanding the applied loads. The mock-up configuration shall simulate a 30-inch suspension system.

b. The pylon swaybraces shall be adjustable and shall have a pad diameter of 1.9 inches.

c. The pylon swaybrace angle shall be 20.7 degrees from the vertical plane through the longitudinal axis of the bomb.

d. The pylon swaybraces shall be located on 20-inch centers midway between the suspension lugs on the bomb.

e. The pylon swaybraces shall be torqued to 290 inch pounds (based on 5/8-18NF-3 threads) for all test loads.

f. Equipment capable of applying a force of 39,000 pounds shall be attached to the test fixture frame in such a manner as to permit the imposition of a downward vertical force, through a suitable load-measuring device, to a pressure pad of 7.4 square-inch area at station 77.5.

g. Equipment capable of applying 27,000 pounds force at each of two stations, simultaneously, shall be attached to the test fixture frame in such a manner as to permit the imposition of downward vertical forces, through suitable load-measuring devices, to pressure pads of 7.4 square-inch area at stations 56.5 and 92.5.

4.6.4.3.2 Test procedures.

4.6.4.3.2.1 Part I. An ejection force of 26,000 pounds shall be applied at station 77.5 in steps of 0, 50, 66.7, 75, 85, 90, 95 and 100 percent increments. The load shall be held at percent increments for three minutes, then relieved to zero load, and the bomb inspected before proceeding to the next percent increment load. The rate of loading shall not exceed 5 percent per second of the 26,000 pounds. After ability of the bomb to withstand 100 percent of the 26,000 pounds has been substantiated, the bomb shall be subjected to 150 percent of the 26,000 pounds in the same percent increments. Any permanent deformation during loading up to and including 115 percent of the 26,000 pounds, or any breakage during loading up to and including 150 percent of the 26,000 pounds shall constitute failure to pass the test.

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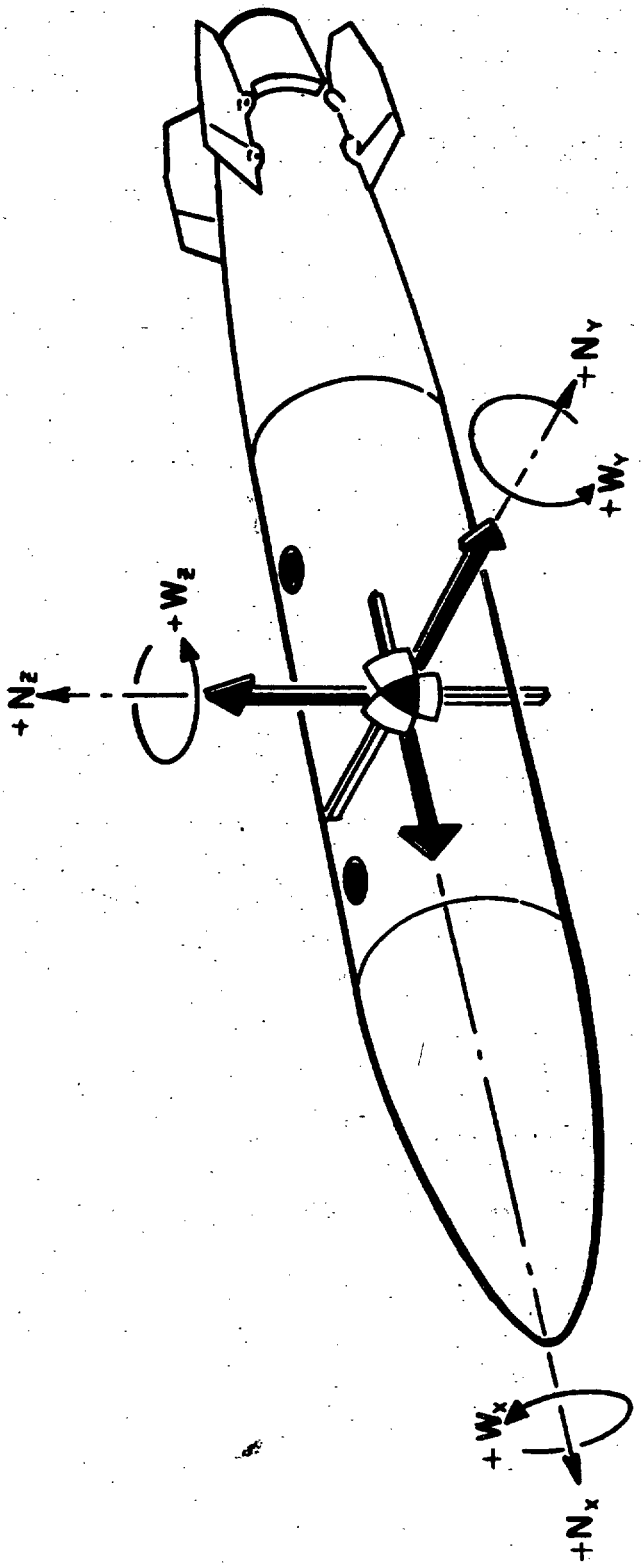


FIGURE 6. REFERENCE AXES AND POSITIVE DIRECTIONS OF AXES.

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4.6.4.3.2.2 Part II. The procedure for Part II shall be identical to that for Part I, except that the ejection force shall consist of two forces of 18,000 pounds each applied at stations 56.5 and 92.5. Rejection criteria shall be the same as for Part I.

4.6.4.4 Fore and aft moment, Parts I and II.

4.6.4.4.1 Part I. With the nose section removed, the forward ballast connection at station 46 shall be tested by applying a positive vertical force of 6560 pounds at station 42.0. This force shall be applied in steps of 0, 50, 66.7, 75, 85, 90, 95 and 100 percent increments. The load shall be held at each percent increment for three minutes, then relieved to zero load, and the bomb inspected before proceeding to the next percent increment load. The rate of loading shall not exceed 5 percent per second of the 6560 pounds force. After ability of the bomb to withstand 100 percent of the 6560 pounds has been substantiated, the test shall be repeated using the same percent loading increments of 150 percent of the 6560 pounds. Any permanent deformation during loading up to and including 115 percent of the 6560 pounds, or any breakage during loading up to 150 percent of the 6560 pounds shall constitute failure to pass the test.

4.6.4.4.2 Part II. The tail cone attachment at station 105 shall be tested by applying a positive vertical force of 5600 pounds at station 135. The method of applying the load, and rejection criteria shall be the same as in Part I.

4.6.4.5 Tail fin test.

4.6.4.5.1 Test set-up. The aft section assembly shall be attached to a rigid frame capable of withstanding the applied loads. The fins shall be in the X configuration with one opposite pair in the horizontal plane. Suitable means shall be provided for measuring fin deflection in the direction of application of loads.

4.6.4.5.2 Test procedure. Loads shall be applied simultaneously to the top side of each of the two opposite fins at the center of pressure of each fin, as indicated on Figure 7. The test shall be performed in the following sequence:

a. Preload the two fins to 400 pounds, and establish a zero reference.

b. Increase the test load directly to 600 pounds.

c. Increase the load in increments of 75 pounds to a total of 1500 pounds. Fin deflection shall be measured at each 75 pound increment up to and including 1500 pounds.

d. Reduce the loads simultaneously to 400 pounds.

e. Determine whether permanent deformation exists.

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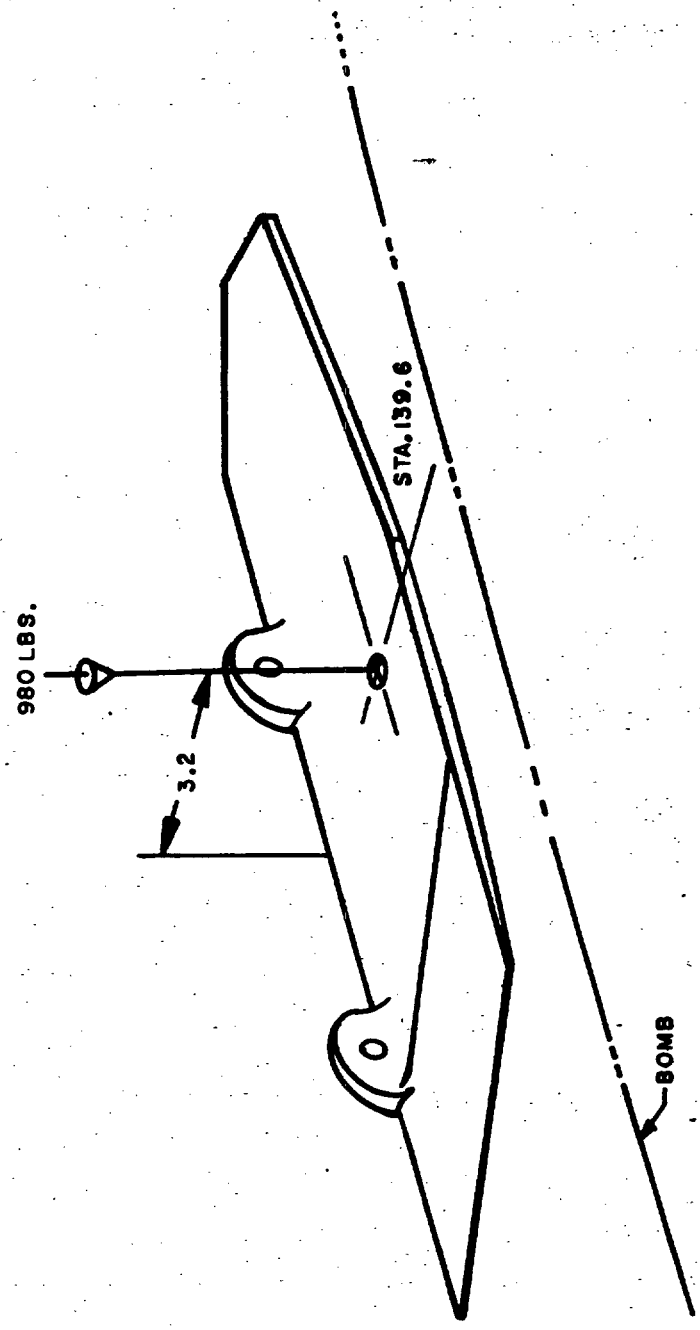


FIGURE 7. APPLICATION OF CONCENTRATED LOADS.

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f. Increase the load directly to 1690 pounds.

g. Increase the load by increments of 50 pounds to a total of 2190 pounds. Fin deflection shall be measured at each 50 pound increment.

h. Reduce loads simultaneously to 400 pounds.

i. Determine whether permanent deformation exists, or failure occurs.

Permanent deformation after loading to 1500 pounds, or failure while loading to 2190 pounds shall constitute failure to pass the test.

4.6.5 Shock test.

4.6.5.1 Test specimen preparation. Bombs to be subjected to the shock test shall be filled with concrete in accordance with the requirements of this specification and MIL-W-27000.

4.6.5.2 Test fixture and test procedure. Each test fixture and test procedure shall be in accordance with MIL-W-27000.

4.6.6 Vibration test.

4.6.6.1 Test specimen preparation. Bombs to be subjected to the vibration test shall be filled with concrete in accordance with the requirements of this specification and MIL-W-27000.

4.6.6.2 Test fixture and test procedure. The test fixture and test procedure shall be in accordance with MIL-W-27000.

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Cleaning all levels. Cleaning and drying shall be in accordance with applicable processes and procedures of MIL-P-116.

5.1.2 Preservation, all levels.

5.1.2.1 Assembled bombs. All bare machined surfaces and surfaces not having a protective finish shall be preserved in accordance with Method I, MIL-P-116.

5.2 Levels of packing.

5.2.1 Level A. Assembled bombs shall be packed in a crate; type 1, class 3, style A, MIL-C-104. The fins and attaching hardware shall be removed from the bomb and suitably secured inside the crate.

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5.2.2 Levels B and C. The assembled bombs shall be packed in a crate conforming to PPP-C-650, type V, style A. The fins and attaching hardware shall be removed from the bomb, packaged in a weather resistant box conforming to PPP-B-636, and secured inside of the crate.

5.3 Physical protection. Cushioning, anchoring, bracing, and blocking, as required, shall be in accordance with MIL-STD-1186. Rough handling tests shall be required only when specified in the contract or order.

5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

BOMB LOADING PRACTICE, BDU-35/E

or

BOMB LOADING PRACTICE, BDU-6/E

6. NOTES

6.1 Intended use. The Bomb, Loading Practice is intended to be used by ground crews to gain practice in ground handling and loading of the parent weapon into types of aircraft which are intended as carriers of the parent weapon.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be conducted by the procuring activity.

c. Levels of preservation, packaging and packing.

6.2.1 Preproduction test samples. It is expected that the contract or purchase order will specify that two bombs will be required as first articles and that these first articles will be subjected to the preproduction tests to determine compliance with the requirements of this specification. It is expected that the contract or purchase order shall specify the point of inspection for these tests.

6.3 Preproduction delivery. No production item should be delivered prior to approval of the preproduction samples.

6.4 Sources. When requesting publications, refer to both title and number.

6.5 Definitions.

a. Preproduction tests. Preproduction tests are those tests performed prior to production on representative samples of the production bomb to demonstrate that the equipment meets the requirements of this specification.

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b. Acceptance tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the bombs are equivalent in performance and construction to the approved preproduction samples.

6.6 Identification of changes. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the change.

Custodian:
Air Force - 79

Reviewer:
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Preparing Activity:
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